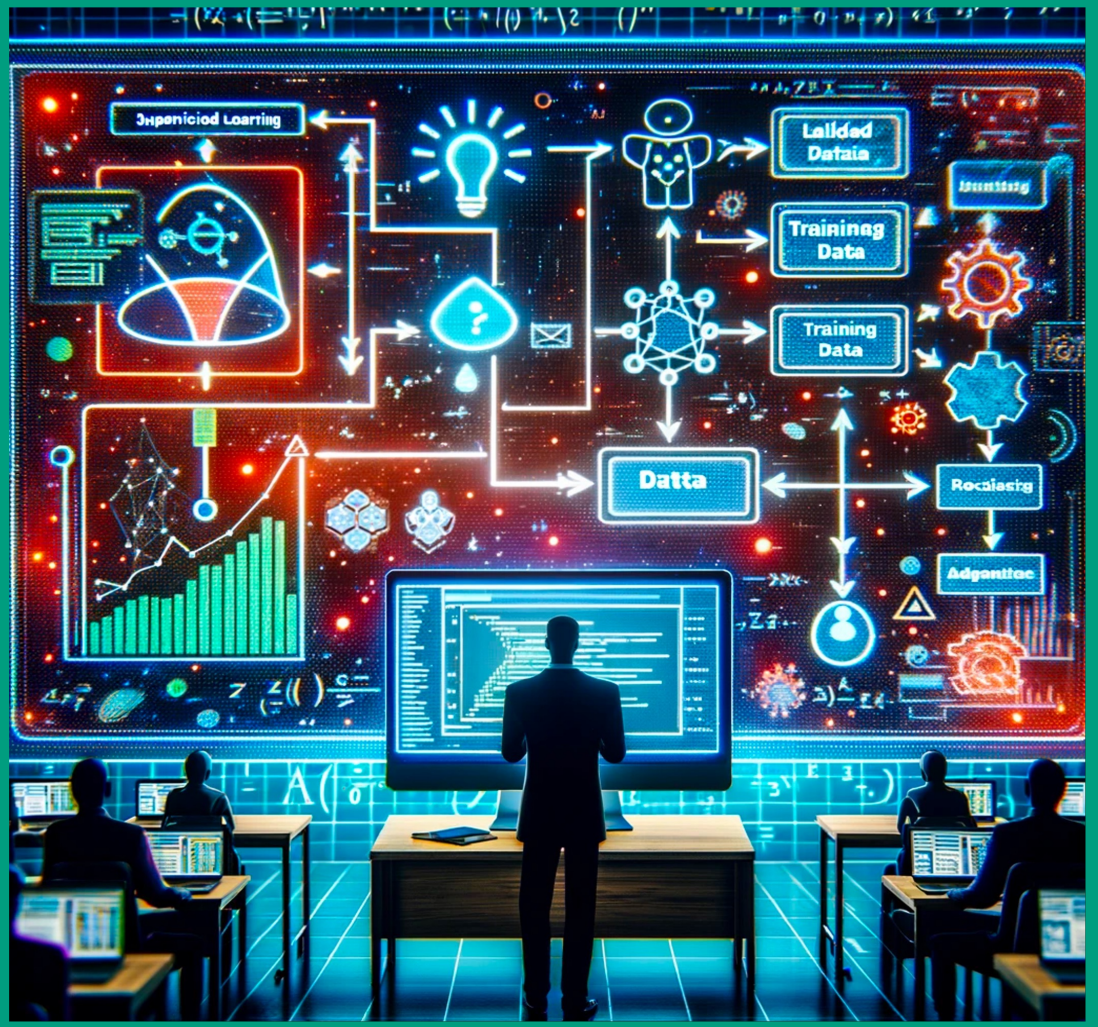
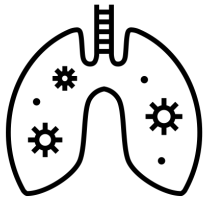
DAT200 - Applied Machine Learning

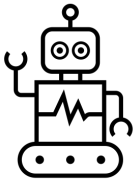
Kristin Tøndel

Lots of data!

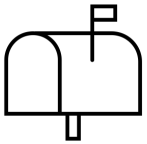




Health care



Robotics



Spam filtering

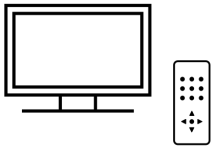




Text analysis (historians, lawyers, social scientists,…)



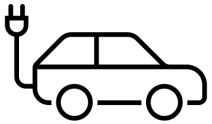
Machine Learning



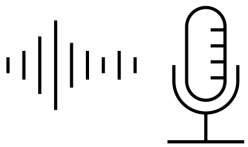
Recommendations



Business analytics



Self-driving vehicles



Speech recognition



Fraud detection



Image recognition (Face detection, Motion detection, Handwriting, Sorting…)

Today’s agenda

Course Organisation

Applied Machine Learning

- What is Machine Learning?
- Three Types of Machine Learning
- Dataset terminology
- Building an ML application



Teachers & TAs

Teacher

- Kristin Tøndel - kristin.tondel@nmbu.no

Teaching Assistants

- Milad Tootoonchi, milad.tootoonchi@nmbu.no
- Lars Johan Tveit Solbakken, lars.johan.tveit.solbakken@nmbu.no
- Liyanapathiranage Sudeepika Wajirakumari Samarathunga, sudeepika.wajirakumari.samarat.liyanapathiranage@nmbu.no

Prerequisites

- Basic data structures
- Basic understanding of probability
- Basic linear algebra
- Basic Python programming

Learning outcomes DAT200

- Overview of common machine learning techniques
- Basic understanding of some ML model properties, operations, algorithms
- Workflow understanding; creating ML pipelines
  - Explorative Analysis, Pre-processing, Modelling, Prediction, Validation, Tuning/Debugging
- Find right ML tools for a given problem
- Domain specific programming skills

Lectures

- Lectures will be live streamed
- Lectures will be recorded
    - > Please help me remember to switch on the microphone!
- Recorded lectures will be available in Canvas «Panopto Video» room shortly after the lecture
- Lecture notes / slides will be made available on Canvas
- Code used in lectures will be made available on Canvas
- Help me make the lectures interactive! Ask questions ;)

Coding in lectures

- Bring your laptop to all lectures with some battery left ☺
- Sometimes I will
  - present code applied to some data set
  - let you do the same analysis on another data set
  - discuss with your neighbour
- Do your coding alone, with the student beside you or in small groups

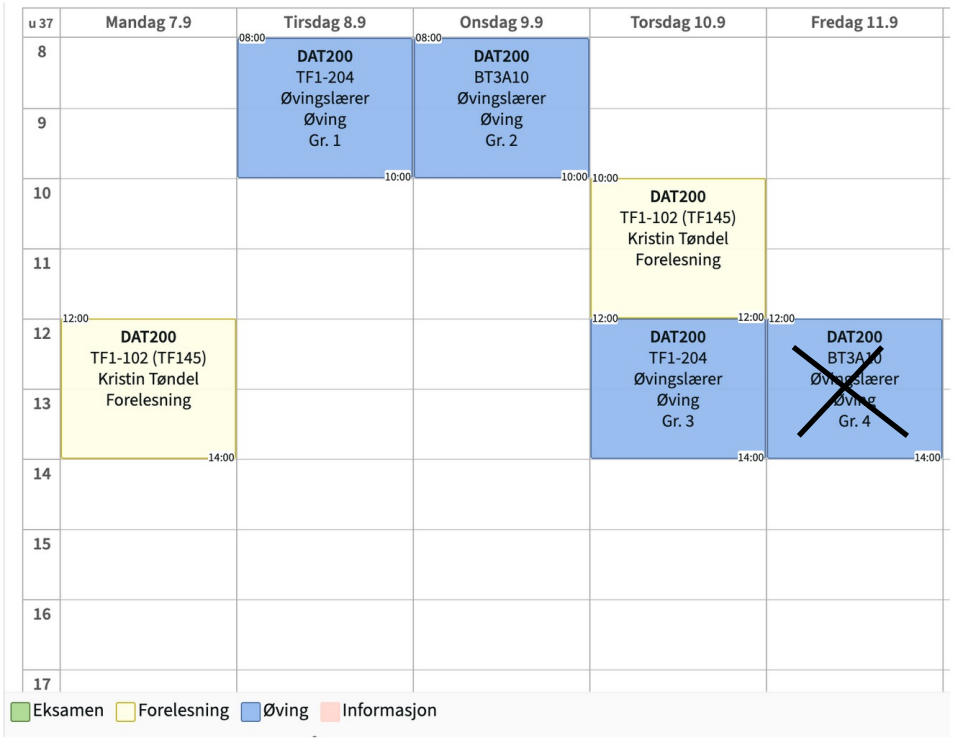
Compulsory assignments (CA)

- Five compulsory assignments in total (CA1, CA2, CA3, CA4, CA5)
- Need to pass **all** compulsory assignments to get admitted to the exam
- 2(3) out of 5 compulsory assignments will include competitions on Kaggle (make a free profile)
- Assignments will be posted in Canvas and will be announced in the lectures

- CA assignments must be submitted in **groups of three students**! The deadline for signing up to an **exercise group** is **Monday September 7th**.
- Respect submission deadlines! TAs give feedback (passed, failed, resubmit improved CA)
- Accepted reasons for delayed submissions
  - Sickness (must provide a certificate by a medical professional) – please let me know before the submission deadline (not after)
  - Study related activities (like excursion, etc.) – need to have information and certificate on this at the start of the semester

Exercises

- At least one TA for each session
- Tentative submission deadlines are posted on Canvas



Getting help

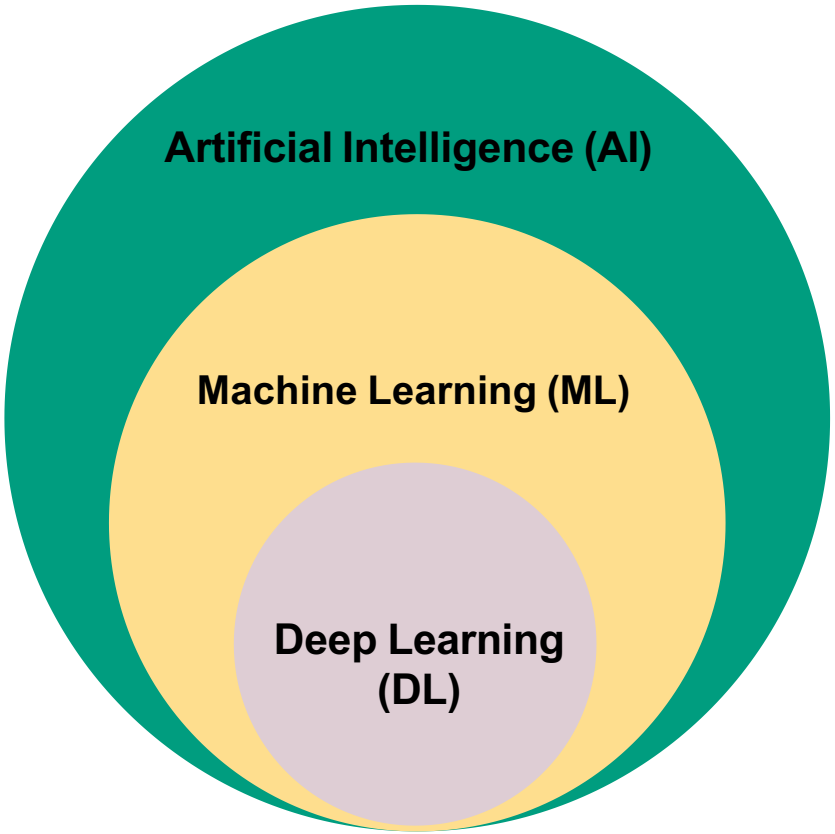
- Go to the exercises, ask TAs
- Padlet (https://padlet.com/kristintondel/dat200-h26-v9q1dhgfell2nve)
- Ask in the lectures (remind me to repeat the question for the recording)
- Use online documentation pages of libraries
- Use the internet! (Search engines, StackOverflow, …)
- Email (if personal matter, otherwise use public channels)

Poll

What is your study program/specialisation?

What is Machine learning?

What is machine learning?



AI: Can we make machines (or do they exist) that can reach human (animal) intelligence? How can we make them?
Narrow AI: Beat/as good as humans in a single task

ML: Giving machines the ability to learn

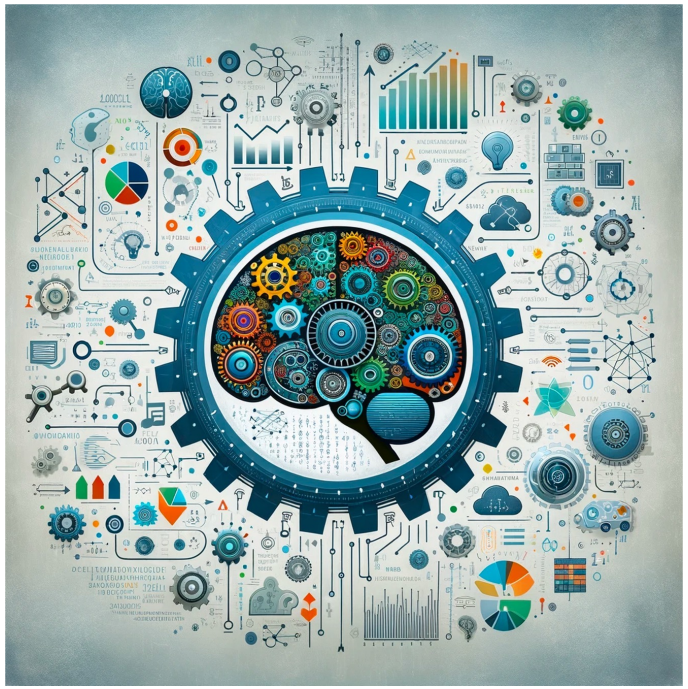
DL: Subfield of ML using models with many parameters (e.g. Deep Neural Networks)

Deep Learning → DAT300

What is machine learning?



Machine learning is the “field of study that gives computers the ability to learn without being explicitly programmed” (Arthur Samuel, 1959)

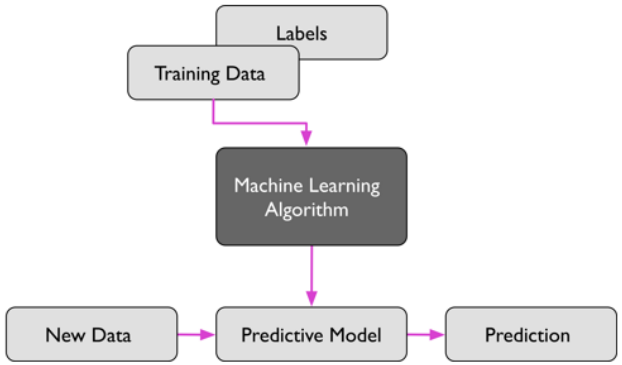


“A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E.” (Tom Mitchell, 1997)

The three paradigms of machine learning

Supervised learning

Supervised learning
- Making predictions about the future



- Main goal in supervised learning
  - **Learn a model** from labeled training data
  - Use the model to **make predictions** on unseen or future data
  - Term «Supervised» refers to a set of samples where the desired **output signals** (labels) **are already known**

Image: S. Raschka, V. Mirjalili. 2017. «Python Machine Learning», Chapter 1, page 3

Supervised Learning

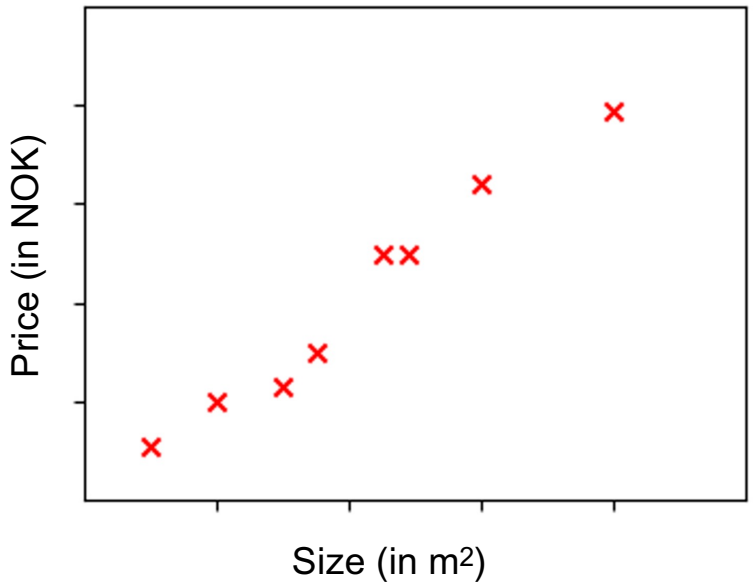
(X, Y)

↓

ML algorithm

↓

M: X → Y



Supervised Learning

Input Labels

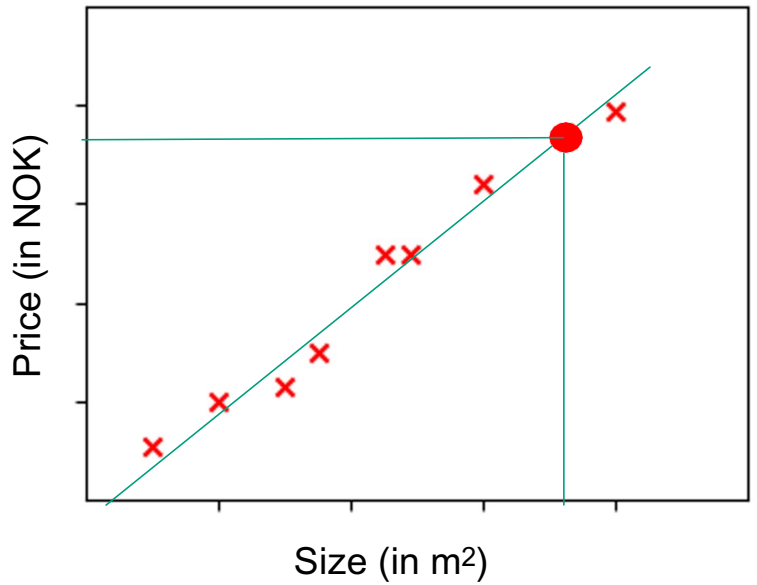
(X, Y)

↓

ML algorithm

↓

$M: X \rightarrow Y$

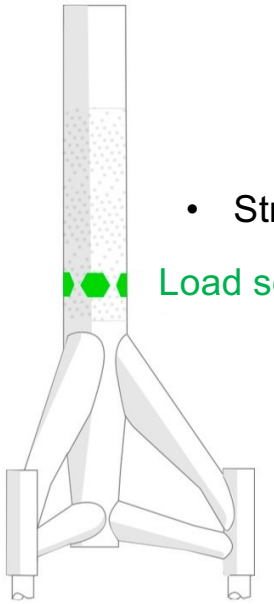


Regression:

Predict a continuous outcome/target/label

Supervised Learning





- Strain (“**Target / Outcome**”, Y)

Standard measurements:
(”**Features**”, X)

- wind speed
- wind direction
- generator output
- turbulence intensity
- pitch angle
- …

→ many dimensions!

e.g. $\boldsymbol{x} = (x_1, x_2, x_3, x_4, x_5)$

$$\boldsymbol{x} \in \mathbb{R}^5$$

What about storms?

Can’t predict completely new situations that have not been seen during training!

Supervised Learning

Input Labels

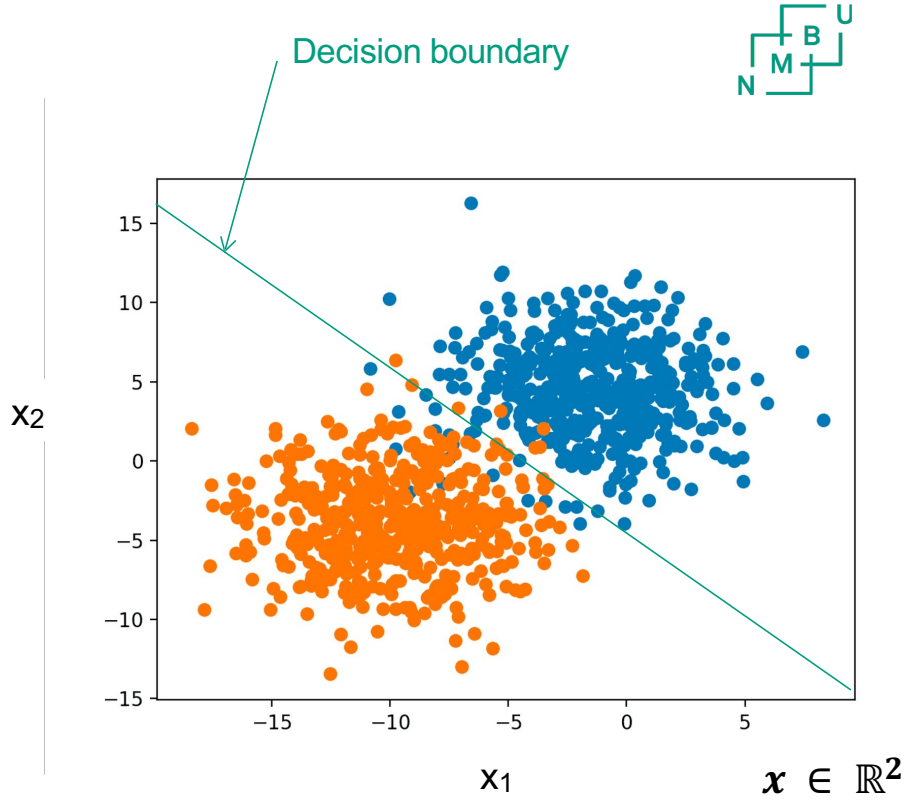
(X, Y)

↓

ML algorithm

↓

$M: X \rightarrow Y$



Classification:

Predict a discrete outcome/target/labels/classes

Supervised Learning

Input



Label

0
1
2
3
4
5
6
7
8
9

Supervised Learning

Input



Label

0
1
2
3
4
5
6
7
8
9

What about?

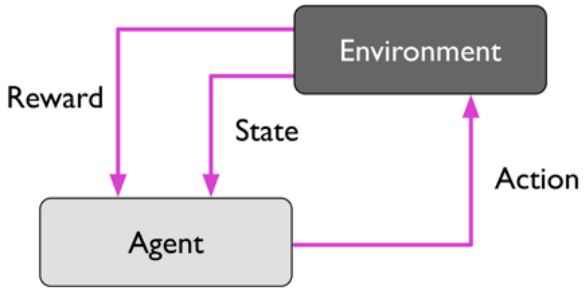


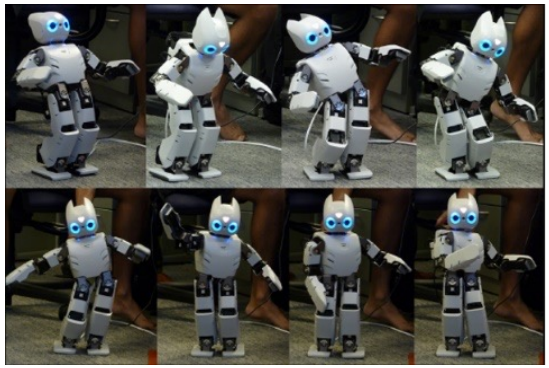
Can’t predict classes that we have not seen during training!

Reinforcement learning

Reinforcement learning

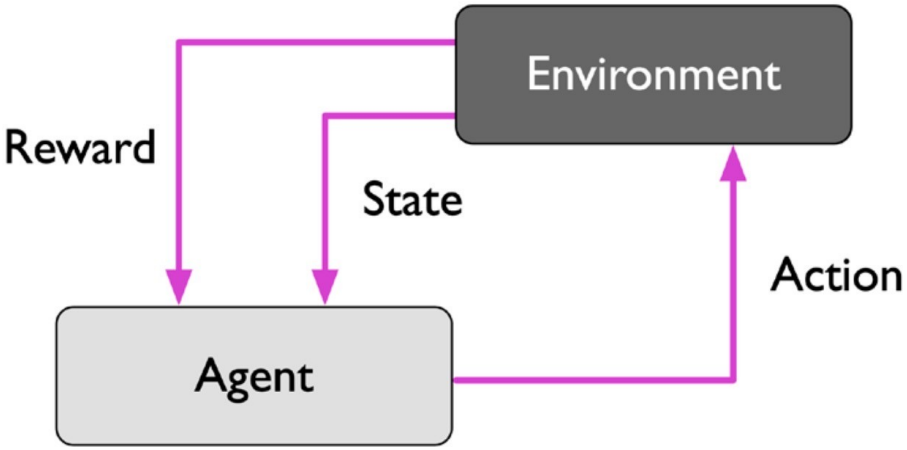
- Solving interactive problems

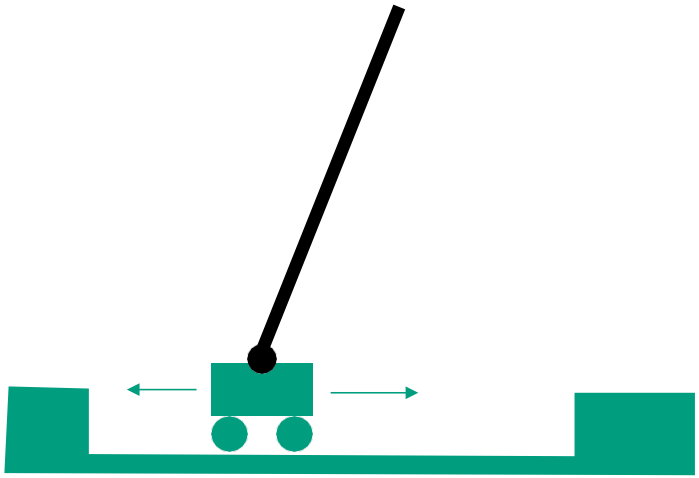




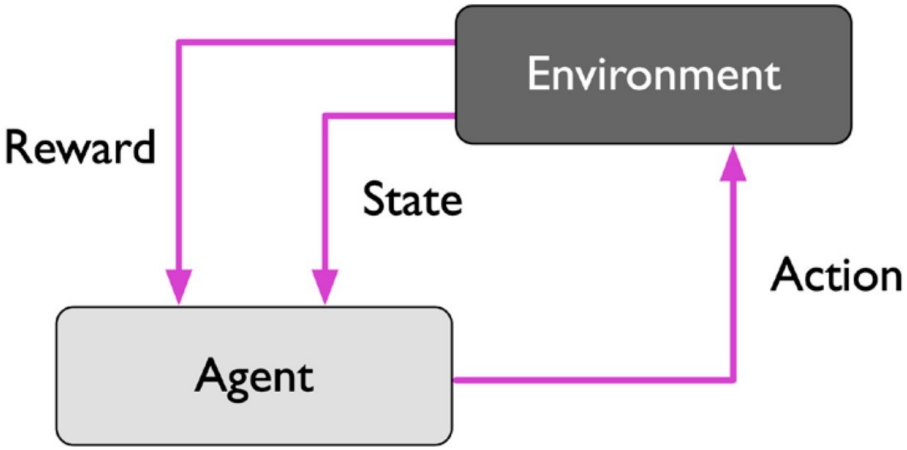
- Goal: Develop a system (**agent**) that **improves performance** based on **interaction** with an **environment**
- Define a **measure** of reward for **particular actions** by the agent
- The agent processes information on the **current state** of the environment
- The state can be associated with a **postive** or **negative** reward
- A reward can be defined as **accomplishing** an **overall goal**
- Concerned with learning a **series of steps** by **maximising a reward** based on
  - Immediate feedback
  - Delayed feedback

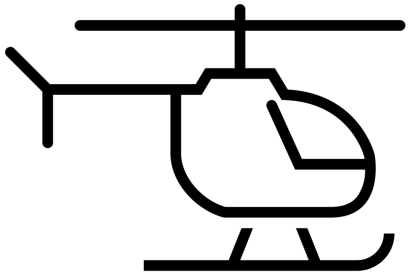
Reinforcement Learning



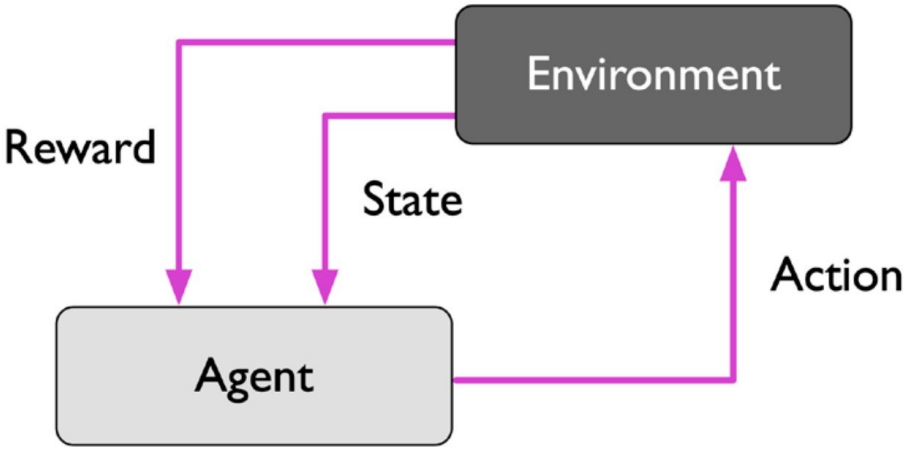


Reinforcement Learning



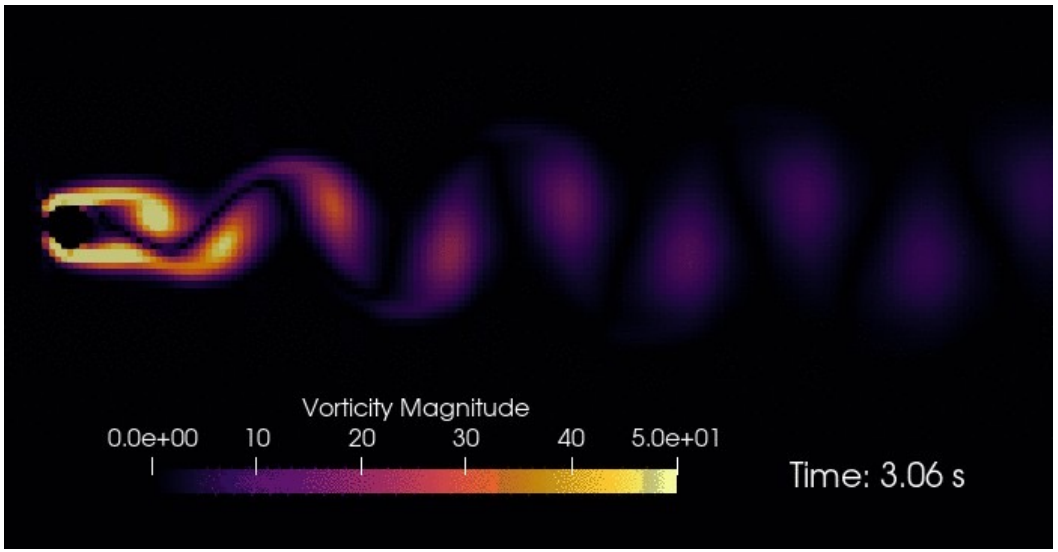


Reinforcement Learning

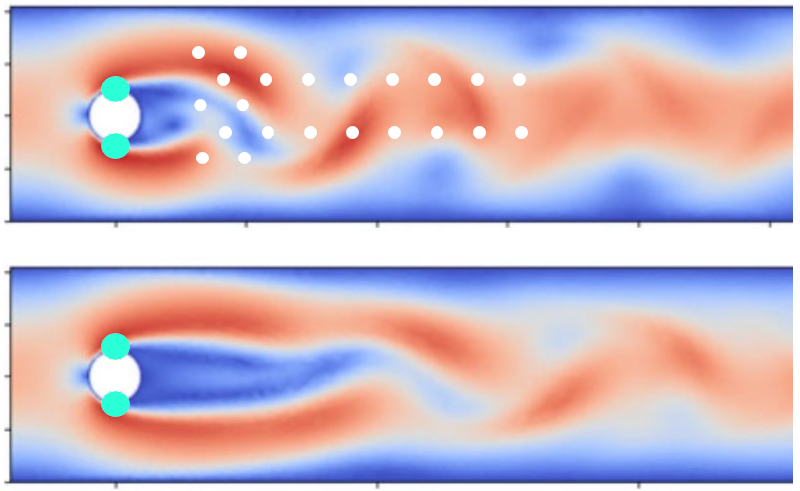




Reinforcement Learning



von Kármán vortex street

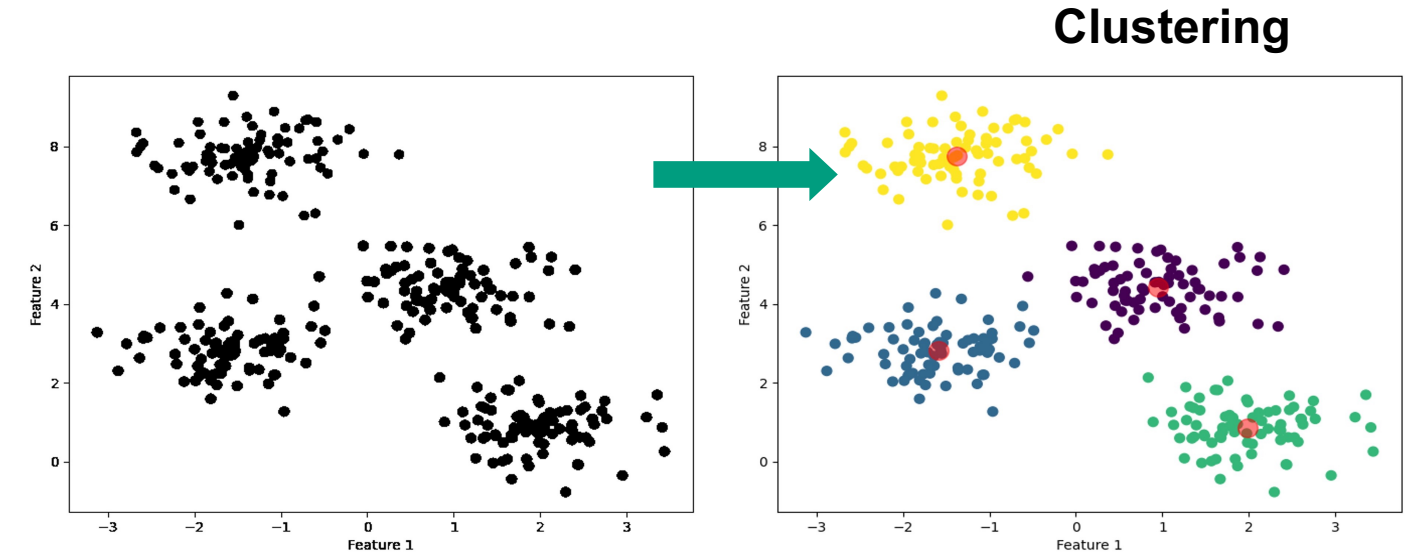


Flow Control; Rabault et al. (2019) *JFM*

Unsupervised learning

Unsupervised Learning

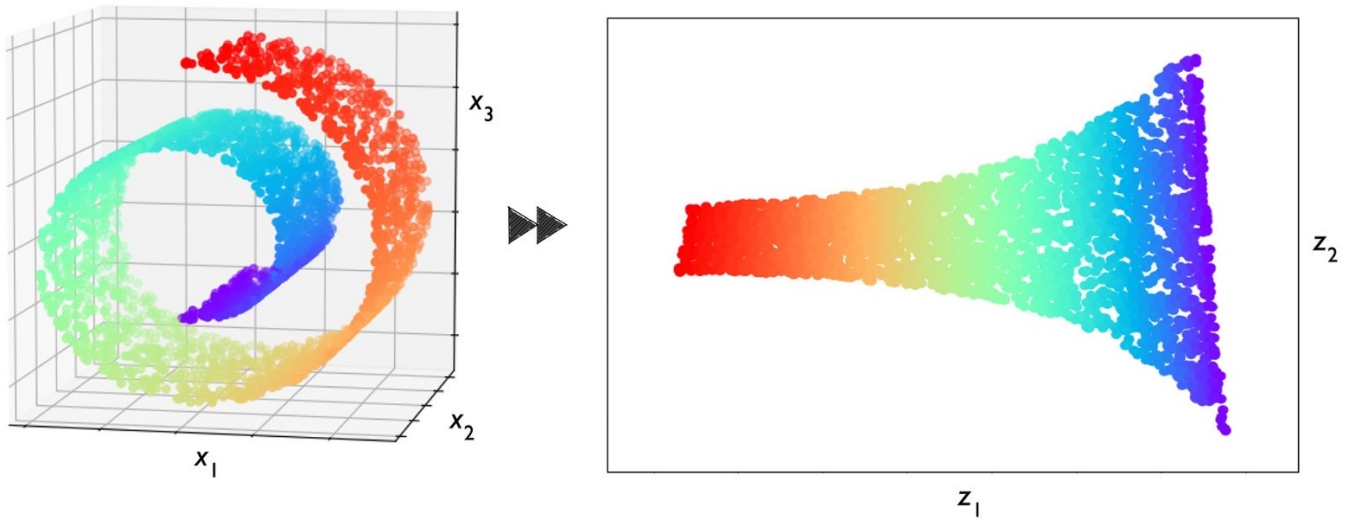
- **Unlabeled** data of **unknown** structure
- **Explore** the structure of the data
  - extract **meaningful** information
  - do so **without guidance** from a known outcome variable or reward function



Unsupervised Learning

Dimension reduction

3D



2D

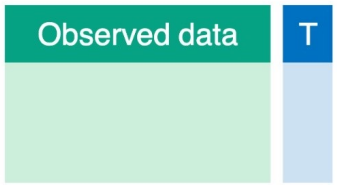
Source: Book Raschka, Liu, Mirjalili, “Machine Learning with PyTorch and Scikit-Learn”

Three types of Machine Learning

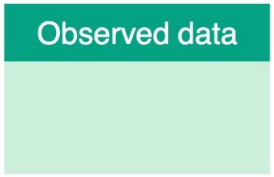
Supervised learning	Labeled data Direct feedback Predict outcome/future
Unsupervised learning	No labels/targets No feedback Find hidden structure in data
Reinforcement learning	Decision process Reward system Learn series of actions

Source: book Raschka, Liu, Mirjalili, “Machine Learning with PyTorch and Scikit-Learn”

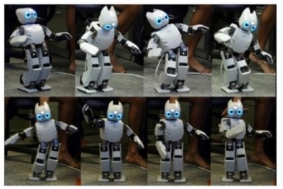
Learning from labelled data



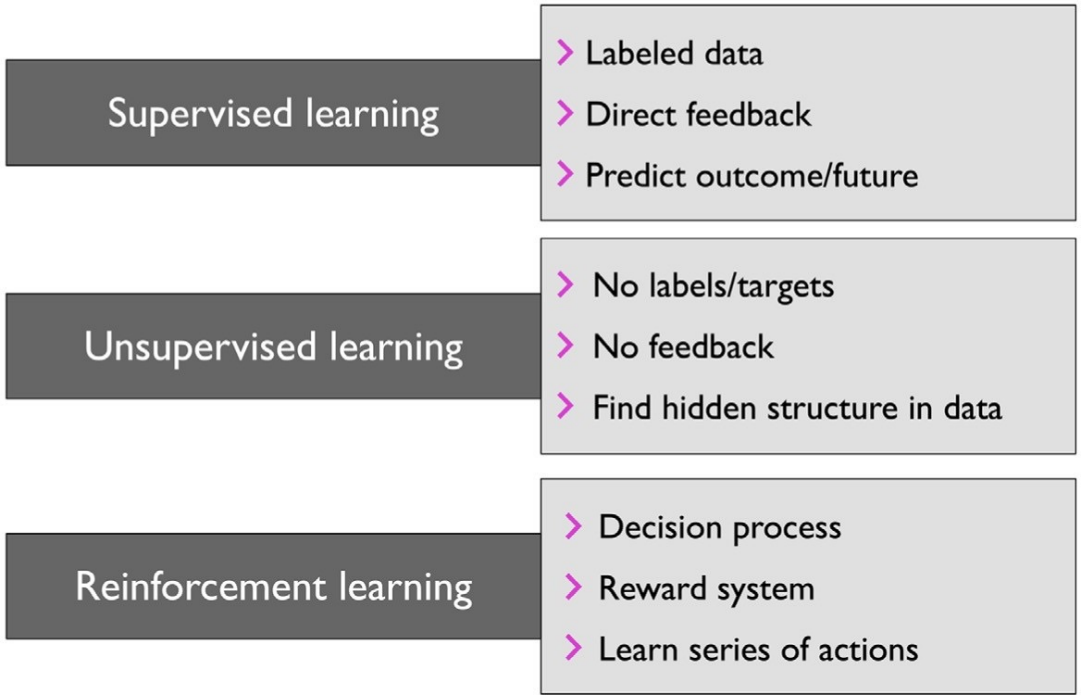
Discover structure in unlabelled data



Learning by “doing” with delayed reward



Three types of Machine Learning



Ch. 2-6, 5, 10

Classification, Regression, Dimension Reduction

Ch. 5, 11

Clustering, Dimension Reduction

Not in DAT200

Tabular data

Tabular data

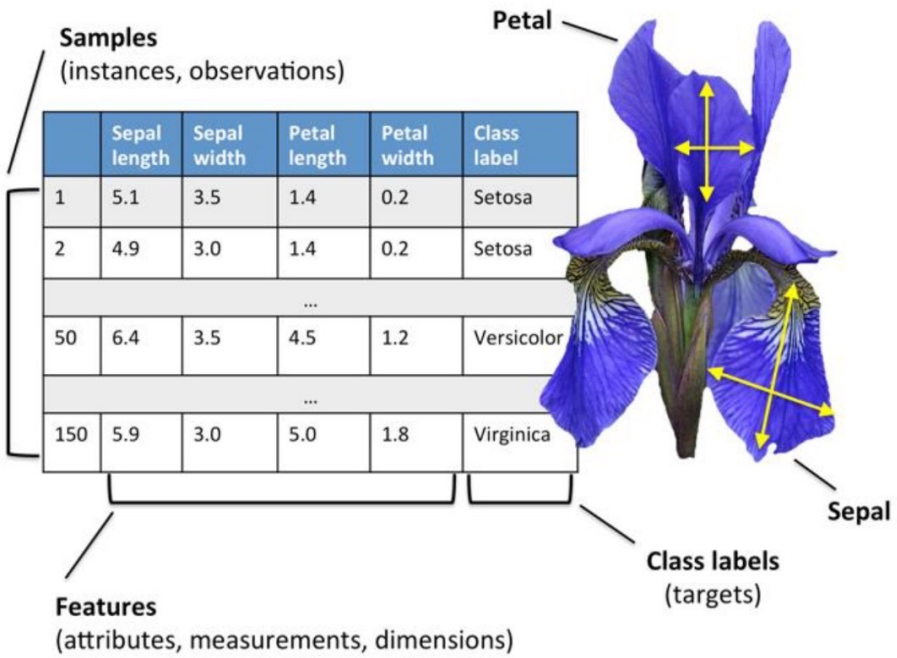
Instances (samples, observations, objects)

Target (class, category, label)

Features (attributes, variables, dimensions)

Patient	Diagnose	Age	Sex	Weight	Comorbidity	Readmitted
1	A45	75	female	85	5	yes
2	L21	45	male	90	1	no

If interested, see: H. Wickham ”Tidy data” (2014) *The Journal of Statistical Software*, vol. 59.



	Sepal length	Sepal width	Petal length	Petal width	Class label
1	5.1	3.5	1.4	0.2	Setosa
2	4.9	3.0	1.4	0.2	Setosa
...					
50	6.4	3.5	4.5	1.2	Versicolor
...					
150	5.9	3.0	5.0	1.8	Virginica

Source: Book Raschka, Liu, Mirjalili, “Machine Learning with PyTorch and Scikit-Learn”

Iris data set

150 **instances** / objects / rows
4 **features** / variables / columns
3 **classes** / targets

Supervised Learning

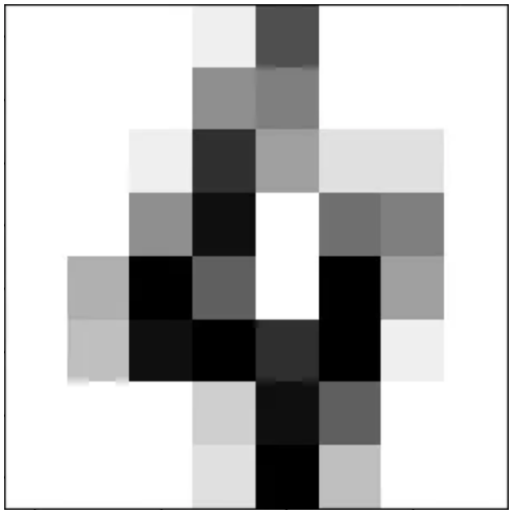
Input



Label

0
1
2
3
4
5
6
7
8
9

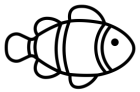
For example:
8*8 “features”
per “sample”



1 label: 4

Be aware

- Datasets we use in DAT200 are unlike real datasets
  - Small
  - Fit in memory
  - Well-structured and organized
  - Nice to analyse (fast results)



- Real world datasets:
  - Huge (may not fit memory nor harddisk)
  - Lots of pre-processing and data cleaning necessary
  - Long process to get full potential



Perfectly OK to learn ML on small datasets!

Basic notation

Vectors and Matrices

We often represent raw (numeric) data as vectors and matrices

Example: Iris data can be represented as a 150 by 4 matrix: $\boldsymbol{X} \in \mathbb{R}^{150x4}$

- Superscript means i-th training sample
- Subscript means j-th feature (dimension)
- Lowercase boldface → vectors ($\boldsymbol{x} \in \mathbb{R}^{nx1}$)
- Uppercase boldface → matrices ($\boldsymbol{X} \in \mathbb{R}^{mxn}$)
- Single element in a vector $x^{(i)}$
- Single element in a matrix $x^{(i)}_{\text{j}}$

$$\begin{bmatrix} x_1^{(1)} & x_2^{(1)} & x_3^{(1)} & x_4^{(1)} \\ x_1^{(2)} & x_2^{(2)} & x_3^{(2)} & x_4^{(2)} \\ \vdots & \vdots & \vdots & \vdots \\ x_1^{(150)} & x_2^{(150)} & x_3^{(150)} & x_4^{(150)} \end{bmatrix}$$

Vectors and Matrices

Row vectors (e.g. one row in $\boldsymbol{X}$, “flower” sample in the Iris data set)

$$\boldsymbol{x}^{(i)} = \begin{bmatrix} x_1^{(i)} & x_2^{(i)} & x_3^{(i)} & x_4^{(i)} \end{bmatrix} \qquad \boldsymbol{x}^{i} \in \mathbb{R}^{1x4}$$

Column vectors (e.g. one column in $\boldsymbol{X}$, one feature)

$$\boldsymbol{x}_j = \begin{bmatrix} x_j^{(1)} \\ x_j^{(2)} \\ \vdots \\ x_j^{(150)} \end{bmatrix} \qquad \boldsymbol{x}_{\mathrm{j}} \in \mathbb{R}^{150x1}$$

Great open-source software!

Machine learning tools in DAT200

Tools



Data & Modeling



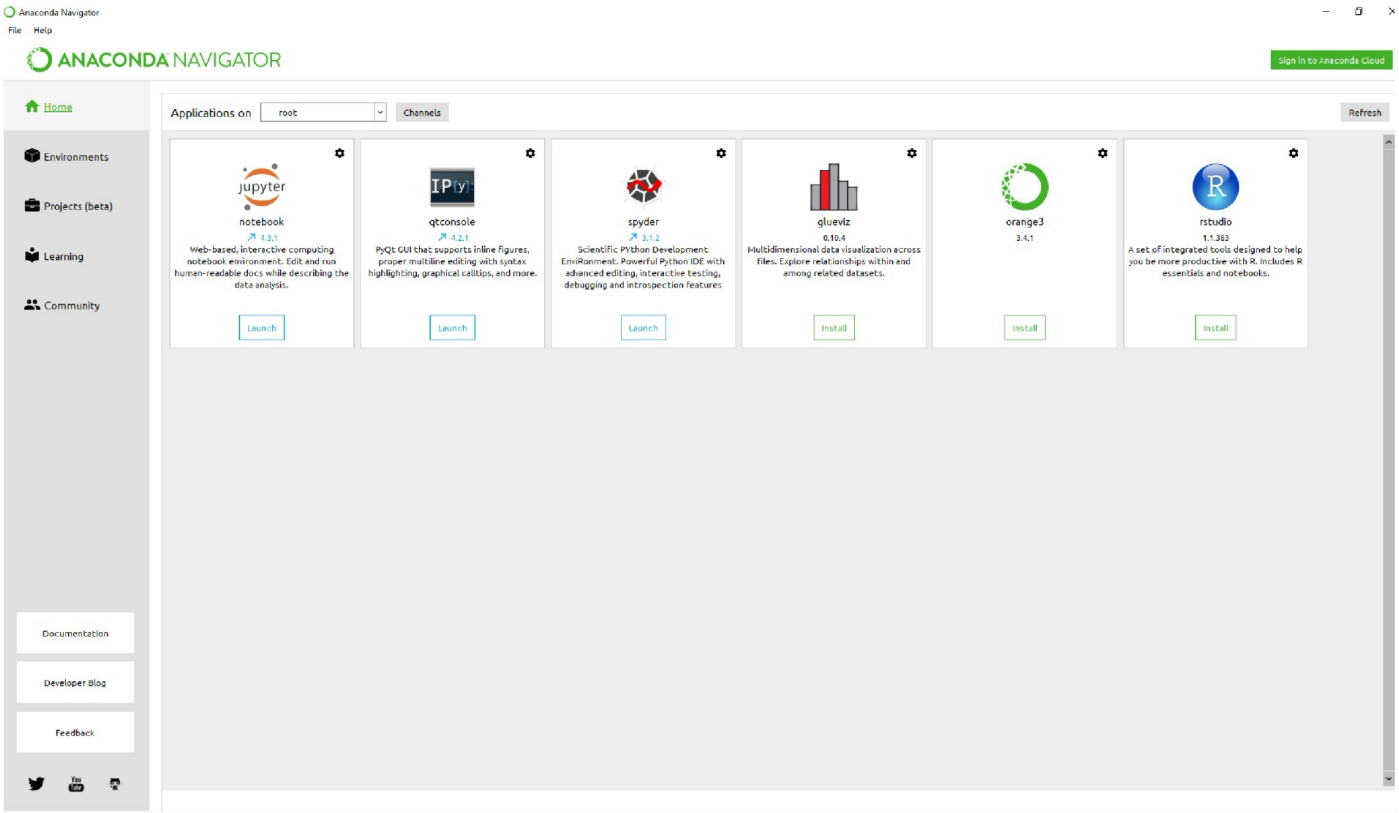
Visualisation



Anaconda navigator

- Convenient working environment
  - Installing and updating various add-on package is straightforward
  - Availability of various useful coding tools
    - Spyder
    - Jupyter notebooks
    - Jupyterlab
    - R + Rstudio
    - Etc.
  - Independent working environments for projects
  - Availability of documentation
  - Links to various communities

Anaconda navigator



Anaconda / Miniconda

- If you want more control, or smaller installation
- Environment file in Canvas: `dat200_environment.yml`
- Use this via the command line

```
conda env create -f dat200_environment.yml

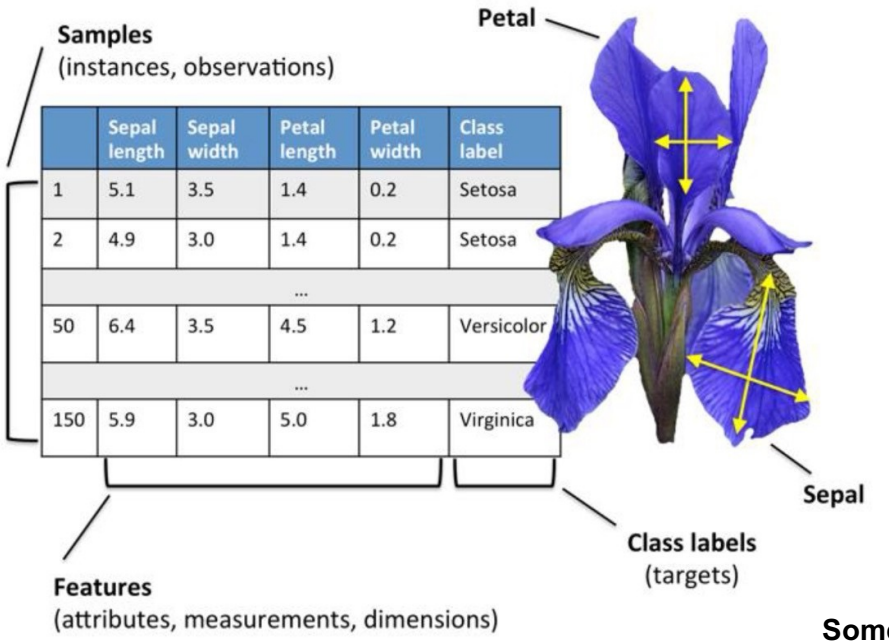
conda activate data200_env

python -m ipykernel install --user --name=dat200_env
```



Pip

- If you know what you are doing
- Use Python 3.10 (for compatibility with provided scripts)
- Install packages as need comes up



	Sepal length	Sepal width	Petal length	Petal width	Class label
1	5.1	3.5	1.4	0.2	Setosa
2	4.9	3.0	1.4	0.2	Setosa
...					
50	6.4	3.5	4.5	1.2	Versicolor
...					
150	5.9	3.0	5.0	1.8	Virginica

Source: Book Raschka, Liu, Mirjalili, "Machine Learning with PyTorch and Scikit-Learn"

Iris data set

150 **instances** / objects / rows
4 **features** / variables / columns
3 **classes** / targets

Some common datasets are shipped with scikit-learn

```
from sklearn.datasets import load_iris

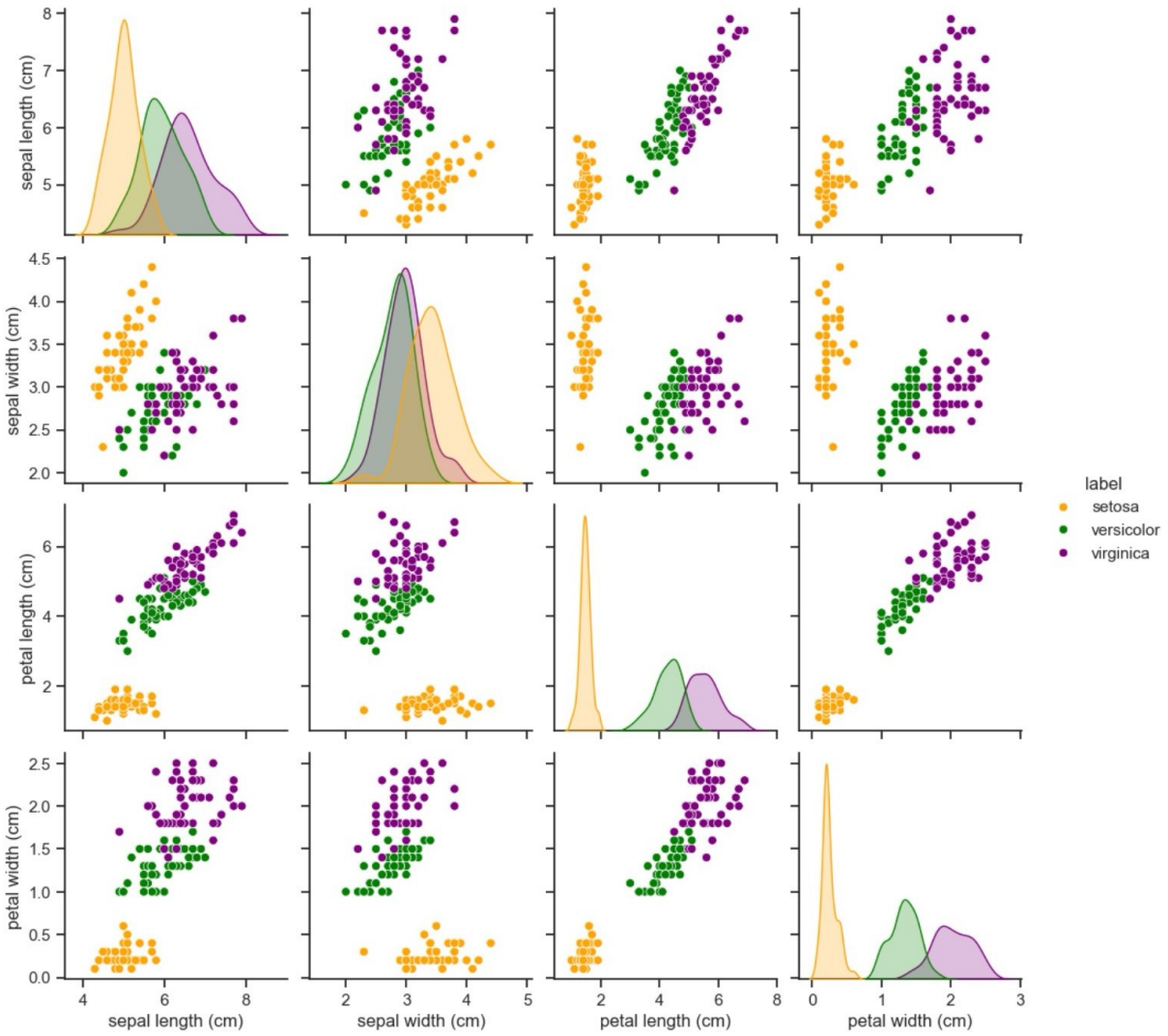
data = load_iris()
dir(data)
print(data.feature_names)
```

Pair plot

- Using Python tooling creating informative plots of the data is quite easy.
- Look in Canvas/Files for “plot_iris.py” for the source code to produce this figure.

```
import seaborn as sns

sns.pairplot
```



Resources

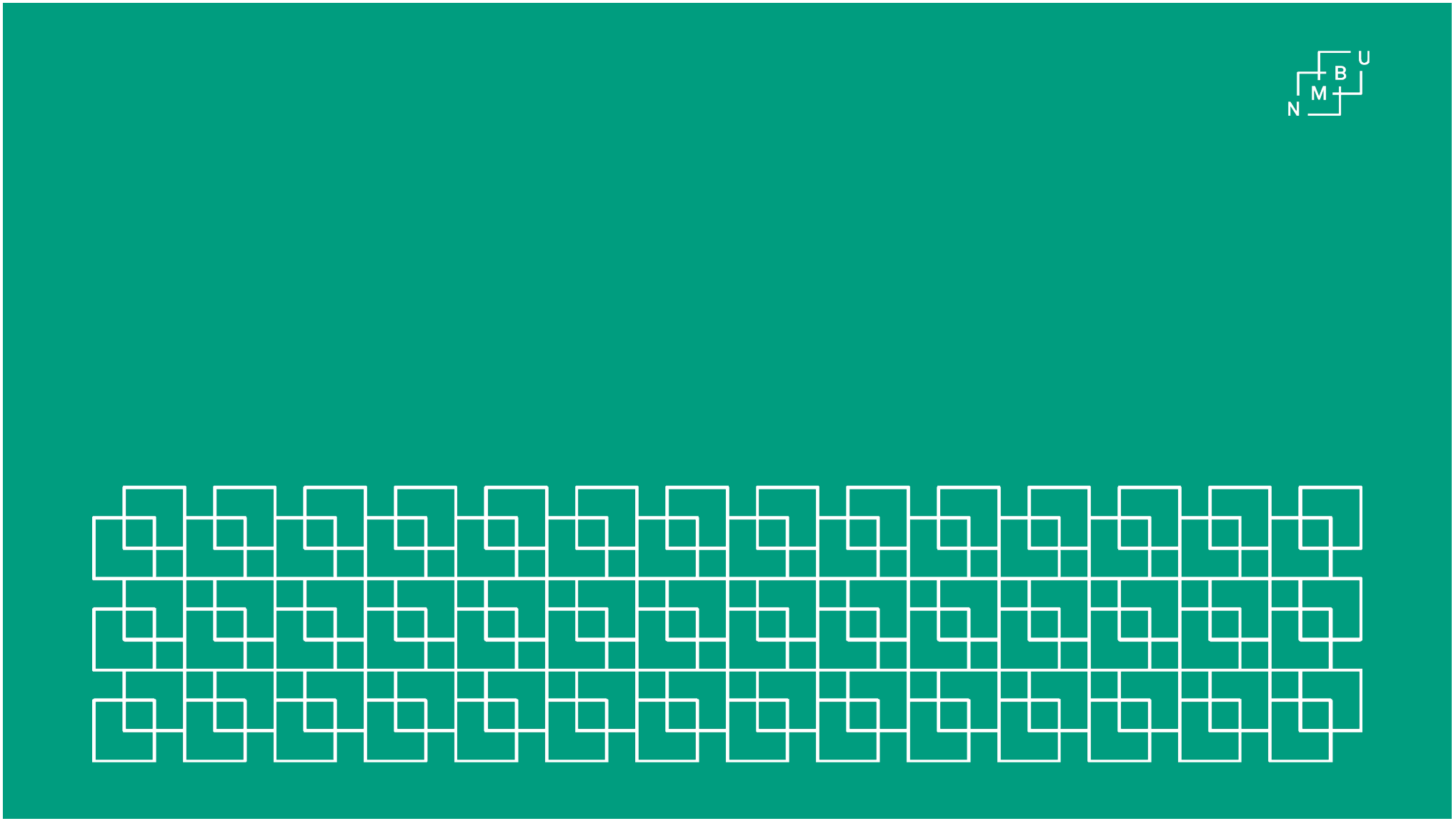
- Python Machine Learning SE, Chapter 1, pages 1 – 16
  - Jupyter notebook: https://github.com/rasbt/python-machine-learning-book-2nd-edition/tree/master/code/ch01

- Anaconda: https://www.anaconda.com/

- scikit-learn: http://scikit-learn.org

- Other scikits: https://scikits.appspot.com/scikits

- CS229 Machine learning, Stanford: http://cs229.stanford.edu/
  - Video lectures: https://www.youtube.com/playlist?list=PLA89DCFA6ADACE599


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